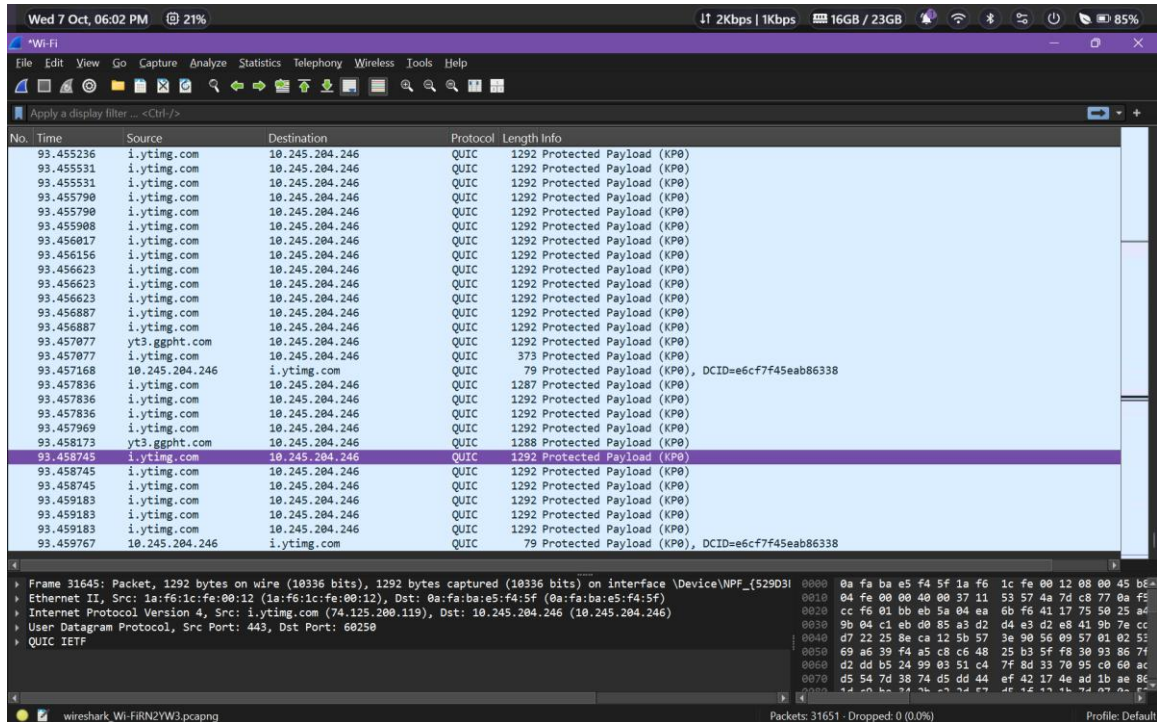


Software and Hardware Tools

Q.1 Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Solution: Network Traffic Capture Using Wireshark

Wireshark is a network packet analyzer used to capture and inspect network traffic.



Top 3 Protocols

No.	Protocol	Purpose
1	TCP	Provides reliable transport of data
2	TLS	Encrypts secure communication
3	DNS	Resolves domain names into IP addresses

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Wi-Fi

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tcp

No.	Time	Source	Destination	Protocol	Length	Info
206	3.976282	10.245.204.246	z-p42-instagram.c10r.insta...	TCP	54	51763 → 443 [ACK] Seq=3145 Ack=24741 Win=65280 Len=0
207	4.005370	z-p42-instagram.c10r...	10.245.204.246	TCP	1329	443 → 51763 [ACK] Seq=24741 Ack=3145 Win=520960 Len=1275 [TCP PDU reassembled in 208]
208	4.005370	z-p42-instagram.c10r...	10.245.204.246	TLSv1.3	1329	Application Data, Application Data
209	4.005584	z-p42-instagram.c10r...	10.245.204.246	TCP	1329	443 → 51763 [ACK] Seq=27291 Ack=3145 Win=520960 Len=1275 [TCP PDU reassembled in 210]
210	4.005584	z-p42-instagram.c10r...	10.245.204.246	TLSv1.3	1329	Application Data
211	4.005767	10.245.204.246	z-p42-instagram.c10r.insta...	TCP	54	51763 → 443 [ACK] Seq=3145 Ack=29841 Win=65280 Len=0
212	4.005903	z-p42-instagram.c10r...	10.245.204.246	TLSv1.3	1329	Application Data
213	4.005903	z-p42-instagram.c10r...	10.245.204.246	TCP	1329	443 → 51763 [PSH, ACK] Seq=31116 Ack=3145 Win=520960 Len=1275 [TCP PDU reassembled in 215]
214	4.012732	z-p42-instagram.c10r...	10.245.204.246	TCP	1329	443 → 51763 [ACK] Seq=32391 Ack=3145 Win=520960 Len=1275 [TCP PDU reassembled in 215]
215	4.012732	z-p42-instagram.c10r...	10.245.204.246	TLSv1.3	1329	Application Data
216	4.013514	10.245.204.246	z-p42-instagram.c10r.insta...	TCP	54	51763 → 443 [ACK] Seq=3145 Ack=34941 Win=65280 Len=0
217	4.017425	z-p42-instagram.c10r...	10.245.204.246	TCP	1329	443 → 51763 [ACK] Seq=34941 Ack=3145 Win=520960 Len=1275 [TCP PDU reassembled in 219]
218	4.017425	z-p42-instagram.c10r...	10.245.204.246	TCP	1329	443 → 51763 [PSH, ACK] Seq=36216 Ack=3145 Win=520960 Len=1275 [TCP PDU reassembled in 219]
219	4.019664	z-p42-instagram.c10r...	10.245.204.246	TLSv1.3	1329	Application Data
220	4.019664	z-p42-instagram.c10r...	10.245.204.246	TCP	1329	443 → 51763 [PSH, ACK] Seq=38766 Ack=3145 Win=520960 Len=1275 [TCP PDU reassembled in 223]
221	4.020010	10.245.204.246	z-p42-instagram.c10r.insta...	TCP	54	51763 → 443 [ACK] Seq=3145 Ack=40041 Win=65280 Len=0
222	4.033090	z-p42-instagram.c10r...	10.245.204.246	TCP	1329	443 → 51763 [ACK] Seq=40041 Ack=3145 Win=520960 Len=1275 [TCP PDU reassembled in 223]
223	4.033090	z-p42-instagram.c10r...	10.245.204.246	TLSv1.3	1329	Application Data
224	4.033090	z-p42-instagram.c10r...	10.245.204.246	TLSv1.3	135	Application Data
231	4.044457	10.245.204.246	lati.ib.opera.technology	TCP	66	21496 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
239	4.060293	10.245.204.246	youtube-ui-1.google.com	TCP	66	3579 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
241	4.062345	lati.ib.opera.techno...	10.245.204.246	TCP	58	443 → 61484 [SYN, ACK] Seq=0 Ack=1 Win=65280 Len=0 MSS=1276
242	4.063228	10.245.204.246	lati.ib.opera.technology	TCP	54	61484 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=0
246	4.067036	10.245.204.246	lati.ib.opera.technology	TCP	1338	61484 → 443 [ACK] Seq=1 Ack=1 Win=65535 Len=1276 [TCP PDU reassembled in 247]
247	4.067036	10.245.204.246	lati.ib.opera.technology	TLSv1.3	816	Client Hello (SHA1+SHA256+op...
248	4.073990	10.245.204.246	z-p42-instagram.c10r.insta...	TCP	54	51763 → 443 [ACK] Seq=3145 Ack=42672 Win=65280 Len=0
268	4.103746	youtube-ui-1.google...	10.245.204.246	TCP	66	443 → 3579 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1276 SACK_PERM WS=256
270	4.104842	10.245.204.246	youtube-ui-1.google.com	TCP	54	3579 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
271	4.104873	10.245.204.246	youtube-ui-1.google.com	TCP	1328	3579 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1276 [TCP PDU reassembled in 271]

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Wi-Fi

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dns

No.	Time	Source	Destination	Protocol	Length	Info
87...	12.077874	10.245.204.246	10.245.204.191	DNS	76	Standard query 0xb8da A www.google.co.in
88...	12.118628	10.245.204.191	10.245.204.246	DNS	101	Standard query response 0xb8da HTTPS www.google.co.in HTTPS
88...	12.130993	10.245.204.191	10.245.204.246	DNS	92	Standard query response 0xb8da A www.google.co.in A 142.251.106.94
98...	12.924819	10.245.204.246	10.245.204.191	DNS	79	Standard query 0x8458 HTTPS graph.instagram.com
98...	12.948871	10.245.204.246	10.245.204.191	DNS	79	Standard query 0x1f9a A graph.instagram.com
98...	12.978369	10.245.204.191	10.245.204.246	DNS	101	Standard query response 0x8458 HTTPS graph.instagram.com CNAME instagram.c10r.instagram.co
98...	13.003509	10.245.204.191	10.245.204.246	DNS	124	Standard query response 0x1f9a A graph.instagram.com CNAME instagram.c10r.instagram.com A
99...	14.948785	10.245.204.246	10.245.204.191	DNS	76	Standard query 0x352a HTTPS www.facebook.com
99...	14.963253	10.245.204.246	10.245.204.191	DNS	76	Standard query 0x9c0f A www.facebook.com
99...	14.987266	10.245.204.191	10.245.204.246	DNS	121	Standard query response 0x9c0f A www.facebook.com CNAME star-mini.c10r.facebook.com A 57.1
99...	14.992144	10.245.204.191	10.245.204.246	DNS	102	Standard query response 0x352a HTTPS www.facebook.com CNAME star-mini.c10r.facebook.com HT
10...	15.942598	10.245.204.246	10.245.204.191	DNS	79	Standard query 0xf98d HTTPS static.xx.fbcdn.net
10...	15.943045	10.245.204.246	10.245.204.191	DNS	79	Standard query 0xf62 A static.xx.fbcdn.net
10...	15.973447	10.245.204.191	10.245.204.246	DNS	183	Standard query response 0xf98d HTTPS static.xx.fbcdn.net CNAME scontent.xx.fbcdn.net HTTPS
10...	15.973447	10.245.204.191	10.245.204.246	DNS	118	Standard query response 0xf62 A static.xx.fbcdn.net CNAME scontent.xx.fbcdn.net A 57.144.
10...	16.489276	10.245.204.246	10.245.204.191	DNS	98	Standard query 0x3d07 PTR 95.134.250.142.in-addr.arpa OPT
10...	16.555335	10.245.204.191	10.245.204.246	DNS	131	Standard query response 0x3d07 PTR 95.134.250.142.in-addr.arpa PTR fx-in-f95.1e100.net OPT
10...	19.113909	10.245.204.246	10.245.204.191	DNS	80	Standard query 0x001d HTTPS accounts.youtube.com
10...	19.157120	10.245.204.246	10.245.204.191	DNS	80	Standard query 0xd5ec A accounts.youtube.com
10...	19.168875	10.245.204.191	10.245.204.246	DNS	158	Standard query response 0x001d HTTPS accounts.youtube.com CNAME www3.l.google.com SOA ns1.
10...	19.196948	10.245.204.191	10.245.204.246	DNS	204	Standard query response 0xd5ec A accounts.youtube.com CNAME www3.l.google.com A 192.178.21
10...	19.542469	10.245.204.246	10.245.204.191	DNS	82	Standard query 0x995e HTTPS static.doubleclick.net
10...	19.542469	10.245.204.246	10.245.204.191	DNS	82	Standard query 0x995e HTTPS static.doubleclick.net

Q.2 Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

Solution: Connectivity Testing Using Ping

```
Wed 7 Oct, 06:08 PM 24%
Administrator: Windows Powe
PS C:\Users\Vaidehi> ping spotify.com

Pinging spotify.com [35.186.224.24] with 32 bytes of data:
Reply from 35.186.224.24: bytes=32 time=43ms TTL=114
Reply from 35.186.224.24: bytes=32 time=54ms TTL=114
Reply from 35.186.224.24: bytes=32 time=63ms TTL=114
Reply from 35.186.224.24: bytes=32 time=23ms TTL=114

Ping statistics for 35.186.224.24:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 63ms, Average = 45ms
PS C:\Users\Vaidehi> ping zomato.com

Pinging zomato.com [18.142.170.124] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 18.142.170.124:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PS C:\Users\Vaidehi> ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [23.55.244.160] with 32 bytes of data:
Reply from 23.55.244.160: bytes=32 time=37ms TTL=55
Reply from 23.55.244.160: bytes=32 time=53ms TTL=55
Reply from 23.55.244.160: bytes=32 time=40ms TTL=55
Reply from 23.55.244.160: bytes=32 time=27ms TTL=55

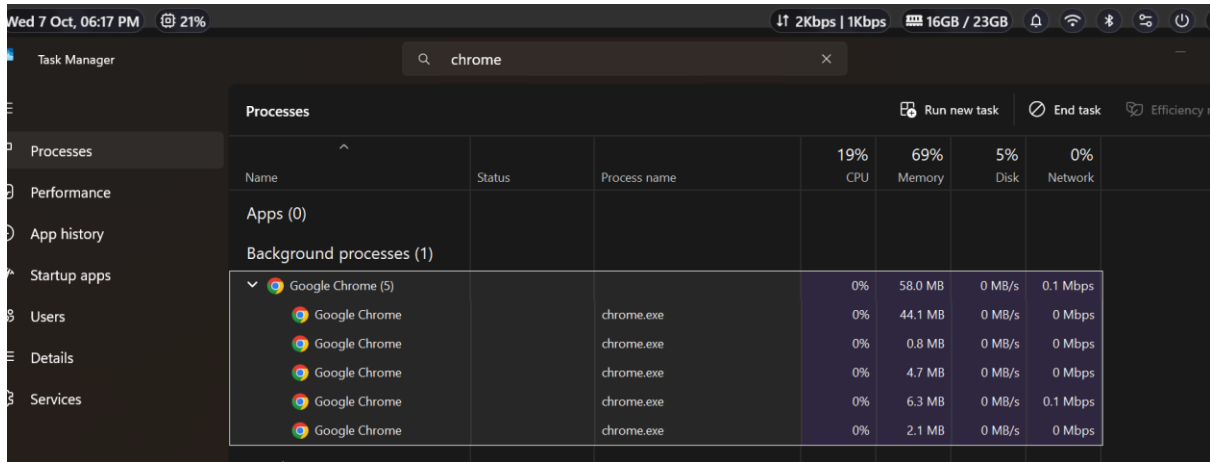
Ping statistics for 23.55.244.160:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 27ms, Maximum = 53ms, Average = 39ms
PS C:\Users\Vaidehi>
```

Comparison

Website	Average Response Time	Result
Spotify	45 ms	Faster
Zomato	39 ms	Slower

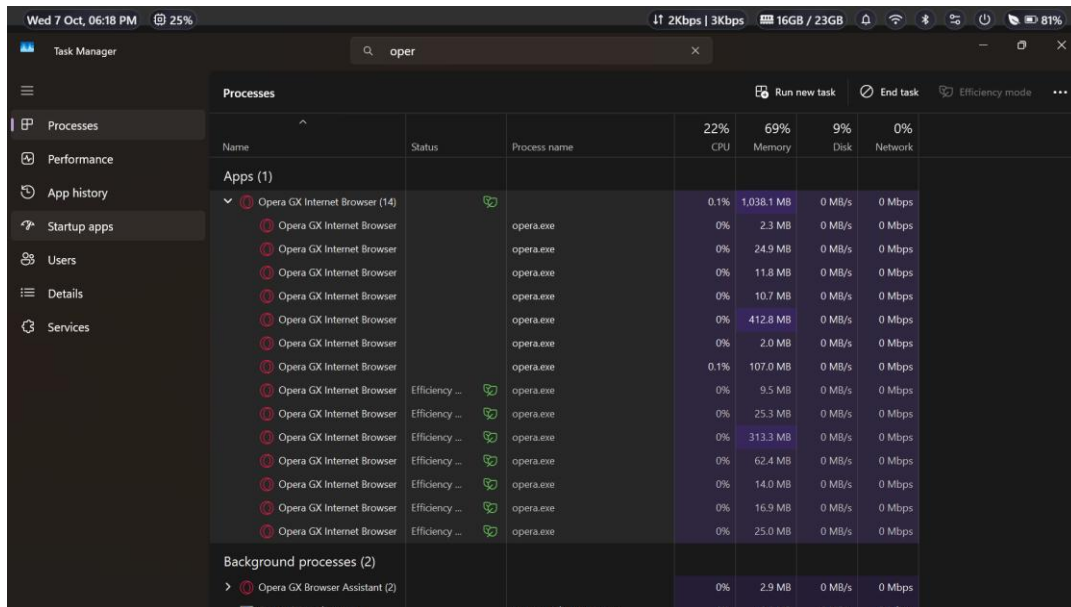
Q.3 Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Solution: Checking Running Processes



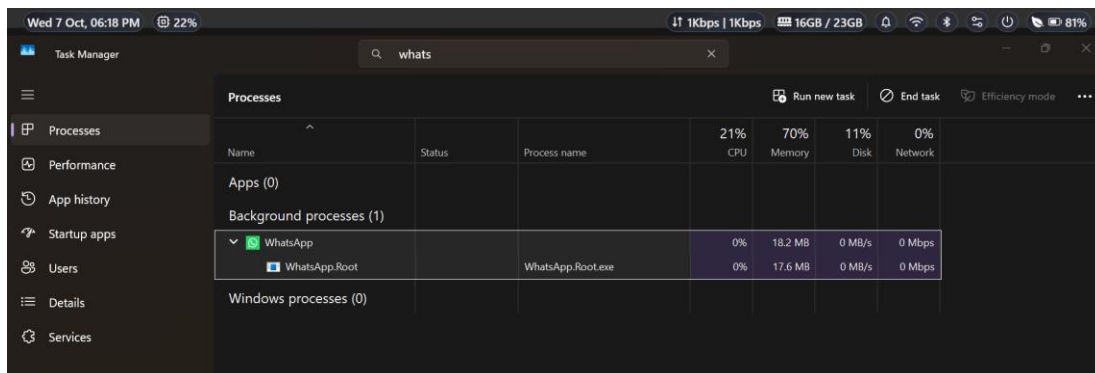
The screenshot shows the Windows Task Manager interface with the search bar set to 'chrome'. The 'Processes' tab is selected, displaying a list of running processes. The table below represents the data shown in the screenshot.

Name	Status	Process name	CPU	Memory	Disk	Network
Apps (0)						
Background processes (1)						
Google Chrome (5)			19%	69%	5%	0%
Google Chrome		chrome.exe	0%	58.0 MB	0 MB/s	0.1 Mbps
Google Chrome		chrome.exe	0%	44.1 MB	0 MB/s	0 Mbps
Google Chrome		chrome.exe	0%	0.8 MB	0 MB/s	0 Mbps
Google Chrome		chrome.exe	0%	4.7 MB	0 MB/s	0 Mbps
Google Chrome		chrome.exe	0%	6.3 MB	0 MB/s	0.1 Mbps
Google Chrome		chrome.exe	0%	2.1 MB	0 MB/s	0 Mbps



The screenshot shows the Windows Task Manager interface with the search bar set to 'oper'. The 'Processes' tab is selected, displaying a list of running processes. The table below represents the data shown in the screenshot.

Name	Status	Process name	CPU	Memory	Disk	Network
Apps (1)						
Opera GX Internet Browser (14)			22%	69%	9%	0%
Opera GX Internet Browser		opera.exe	0.1%	1,038.1 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	2.3 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	24.9 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	11.8 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	10.7 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	412.8 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	2.0 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0.1%	107.0 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	9.5 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	25.3 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	313.3 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	62.4 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	14.0 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	16.9 MB	0 MB/s	0 Mbps
Opera GX Internet Browser		opera.exe	0%	25.0 MB	0 MB/s	0 Mbps
Background processes (2)						
Opera GX Browser Assistant (2)			0%	2.9 MB	0 MB/s	0 Mbps
Opera GX Browser Assistant (2)			0%	0.8 MB	0 MB/s	0 Mbps



The screenshot shows the Windows Task Manager interface with the search bar set to 'whats'. The 'Processes' tab is selected, displaying a list of running processes. The table below represents the data shown in the screenshot.

Name	Status	Process name	CPU	Memory	Disk	Network
Apps (0)						
Background processes (1)						
WhatsApp			21%	70%	11%	0%
WhatsApp.Root		WhatsApp.Root.exe	0%	18.2 MB	0 MB/s	0 Mbps
WhatsApp.Root		WhatsApp.Root.exe	0%	17.6 MB	0 MB/s	0 Mbps
Windows processes (0)						

Q.4 Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Hint : Make sure your firewall allows scanning, and only scan your own home/college network.

Solution: Network Scanning Using Advanced IP Scanner

Advanced IP Scanner

Steps

- 1. Download and install Advanced IP Scanner.**
- 2. Connect the computer to the Wi-Fi network you want to scan.**
- 3. The software will automatically detect the local IP range.**
- 4. Check the detected IP range, for example 192.168.10.0/24.**
- 5. Click Scan and wait for the scan to finish.**
- 6. Check the list of connected devices.**
- 7. Record the IP address and device name of at least two devices.**

Example Result

No.	IP Address	Device
1	192.168.10.1	Wi-Fi Router
2	192.168.10.5	Laptop
3	192.168.10.10	Printer
4	192.168.10.20	Smart TV

Note : These are sample results for demonstration. The actual devices and IP addresses will depend on the network being scanned. Cuz the scan should only be performed on a network that I own or have permission to test.

Q.5 Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Solution: Network Troubleshooting Hardware Tools

1. Ethernet Cable Tester

Use : An Ethernet cable tester is used to check whether an Ethernet cable is wired correctly and whether there are broken or incorrectly connected wires. It is especially useful when a computer or network device cannot establish a wired connection.

2. RJ-45 Crimping Tool

Use : An RJ-45 crimping tool is used to attach an RJ-45 connector to an Ethernet cable and create or repair network cables.

For example, when making a T568B Ethernet cable, the wires must be arranged in the correct pin order before crimping. The Ethernet practical provided earlier shows the T568B pin arrangement and explains the RJ-45 pin layout. Ethernet Standard and Specifica...

3. Digital Multimeter

Use : A digital multimeter can be used to measure electrical voltage, resistance, and continuity when troubleshooting power or hardware-related network problems.

For example, it can help check whether a power adapter or cable is supplying the expected voltage.